

Domain

A function's domain is its set of allowed input values.

What would prevent an input from being allowed?

Range

A function's range is its set of actual output values.

Why is the range not always all real numbers?

A finite relation

$$\{(-2, 5), (0, 5), (3, -1)\}$$

What are the domain and range?

Interval notation

$(-3, 7]$

Which input values does this interval include?

A polynomial's domain

$$f(x) = x^3 - 2x + 4$$

What is its real domain?

A rational function's domain

$$f(x) = \frac{x + 1}{x^2 - 9}$$

What is its domain in interval notation?

Simplification and the domain

$$f(x) = \frac{x^2 - 4}{x - 2}$$

Does cancellation make $x = 2$ an allowed input?

A square root's domain

$$f(x) = \sqrt{7 - 2x}$$

What is its real domain?

A root in a denominator

$$f(x) = \frac{1}{\sqrt{x-3}}$$

Why is the endpoint treated differently from a square root alone?

Two restrictions at once

$$f(x) = \frac{\sqrt{x+2}}{x-1}$$

What inputs satisfy both restrictions?

An odd root's domain

$$f(x) = \sqrt[3]{x + 5}$$

What is its real domain?

A shifted square root

$$f(x) = \sqrt{x - 4} + 2$$

What are its domain and range?

A quadratic's range

$$f(x) = (x - 3)^2 - 5$$

What is its range?

A reciprocal's range

$$f(x) = \frac{1}{x} + 2$$

Which output is impossible?

A graph with endpoints

A graph is the straight segment from $(-2, 3)$ to $(4, -1)$, including both endpoints.

What are its domain and range?

An excluded endpoint

A graph is the straight segment from $(-2, 3)$ to $(4, -1)$, with the left endpoint excluded and the right included.

What are its domain and range?

A contextual domain

A box contains 24 pencils. Let $R(n) = 24 - n$ be the number remaining after n pencils are removed.

What domain fits this situation?

Piecewise functions

A piecewise function uses different rules on different parts of its domain.

How do we decide which rule to use for an input?

Evaluate a piecewise rule

Use $f(x) = x + 2$ when $x < 1$ and $f(x) = 4 - x$ when $x \geq 1$.

$f(1)$

Which rule applies, and what output results?

Sketch a piecewise function

Use $f(x) = x + 2$ when $x < 1$ and $f(x) = 4 - x$ when $x \geq 1$.

What graph represents these two rules and their boundary?

A jump at a boundary

Use $g(x) = 2$ when $x < 0$ and $g(x) = x + 4$ when $x \geq 0$.

What is the range of g ?

A piecewise context

Parking costs \$4 for up to 1 hour and \$7 for more than 1 hour through 2 hours. Use $0 < t \leq 2$.

What piecewise function represents the cost?